Educational Attainment and Environmental Exposure: Assessing the Link Between High School Completion and Toxic Releases in California.*

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This paper explores the relationship between high school educational attainment and toxic emissions from industrial facilities across California census tracts using data from CalEnviroScreen 4.0. Educational attainment is examined as a socioeconomic factor that may influence where residents live in relation to environmental hazards. A simple linear regression indicates that census tracts with lower high school completion tend to experience higher levels of toxic emission, but data issues and violations of regression assumptions cast doubt on the reliability of the results. Simple linear regression may therefore be unsuitable for this research question, and future analyses could apply data transformations or alternative models to better capture the relationship.

1 Introduction

This paper investigates the relationship between high school educational attainment and toxic emissions from industrial facilities across regions in California. Educational attainment is an important socioeconomic factor, often linked to income, social mobility, and health (Breen and Jonsson 2005; Walsemann, Gee, and Ro 2013). While previous studies have examined how socioeconomic factors such as race and wealth relate to residence in areas with higher pollution, the specific role of education has not been fully explored (Liu et al. 2021). Because these dynamics can influence where individuals and communities reside, this study evaluates whether lower educational attainment is associated with greater exposure to toxic waste levels.

*Project repository available at: https://github.com/ekusnadi/Math-261A-Project-1.

Educational attainment and toxic release data from the CalEnviroScreen 4.0 data dashboard were employed to fit a simple linear regression model using R (Office of Environmental Health Hazard Assessment 2021a; R Core Team 2025). While the analysis indicates a negative association between high school educational attainment percentages and toxic waste levels across California census tracts, underlying issues in the data limit the strength of this conclusion.

The structure of the remainder of this paper will be as follows: Section 2 gives an overview of the data, Section 3 explains the methods used in the analysis, and Section 4 describes the results and possible further explorations.

2 Data

Data for this project was obtained from the Office of Environmental Health Hazard Assessment (OEHHA) CalEnviroScreen 4.0 data dashboard (Office of Environmental Health Hazard Assessment 2021b). CalEnviroScreen is a mapping tool for ranking and categorizing California communities based on socioeconomic, environmental, and health data. This data is sourced from various state and federal government organizations, including the California Air Resources Board (CARB), California Department of Pesticide Regulation (CDPR), Department of Toxic Substances Control (DTSC) (Office of Environmental Health Hazard Assessment 2021b).

The raw data is available for download as an Excel spreadsheet from OEHHA's CalEnviroScreen website, with metrics for every individual census tract in California (Office of Environmental Health Hazard Assessment 2021b). A census tract is a geographic subdivision of a county defined for the purpose of collecting and analyzing census data. For this analysis, the relevant columns in this dataset are 'Education' and 'Tox. Release', which corresponds to educational attainment and toxic release levels from industrial facilities.

The educational attainment data represents the percent of the population within a particular census tract that is over age 25 and has less than a high school education. This data was originally sourced from the American Community Survey (ACS), an ongoing survey of the US population conducted by the US Census Bureau (U.S. Census Bureau 2019). These values are based on 5-year ACS estimates from 2015 to 2019. Each census tract estimate was evaluated for reliability using both the standard error (SE) and relative standard error (RSE). Estimates were included in the dataset only if they met at least one reliability criterion: an RSE less than 50 or an SE less than the mean SE of all California census tract estimates for education. Because educational attainment is a percentage, the bounded scale may complicate model fit and interpretation.

The toxic releases from industrial facilities data contains the toxicity-weighted concentrations of modeled chemical releases to air from facility emissions and off-site incineration averaged over 2017 to 2019 and releases from Mexican facilities averaged over 2014 to 2016. This combines data from the Toxics Release Inventory (TRI) and Risk Screening Environmental

Indicators (RSEI) maintained by the U.S. Environmental Protection Agency, as well as the Mexico Registry of Emissions and Transfer of Contaminants (RETC) (U.S. Environmental Protection Agency (EPA) 2021; Secretaría de Medio Ambiente y Recursos Naturales (SEMARNAT) 2021). Southern California borders Mexico, so the RETC data accounts for this cross-border pollution. Annual release of listed chemicals (in lbs/yr) in the TRI dataset is self-reported by facilities annually if they meet a threshold of at least 10 employees and at least 25,000 lbs manufactured or 10,000 lbs used of certain chemicals. RSEI then transforms the TRI and RETC data using chemical-specific toxicity weights and atmospheric dispersion models to produce an estimated pollution concentration score. This score is designed for comparing relative pollution levels across locations but lacks meaningful units and is bounded below by zero, which may limit interpretability in the linear regression.

Although both variables are aggregated at the census tract level, they represent data from slightly different time periods. The educational attainment data are estimates from 2015 to 2019, while the toxic release data are averaged over 2017 to 2019 as well as data from Mexican facilities averaged over 2014 to 2016. This discrepancy in time may lead to minor inconsistencies in the results, as population characteristics or industrial activity could have changed over those years.

All census tracts containing missing values ("NA") for either high school educational attainment or toxic release were removed from the dataset. This resulted in the removal of 103 rows, leaving 7932 observations for analysis. Several outliers are present in the toxic release data, but they are retained in the current model, as their accuracy has not yet been verified (Office of Environmental Health Hazard Assessment 2021b). Figure 1 shows the frequency distributions of the observed values, both of which are heavily right-skewed.

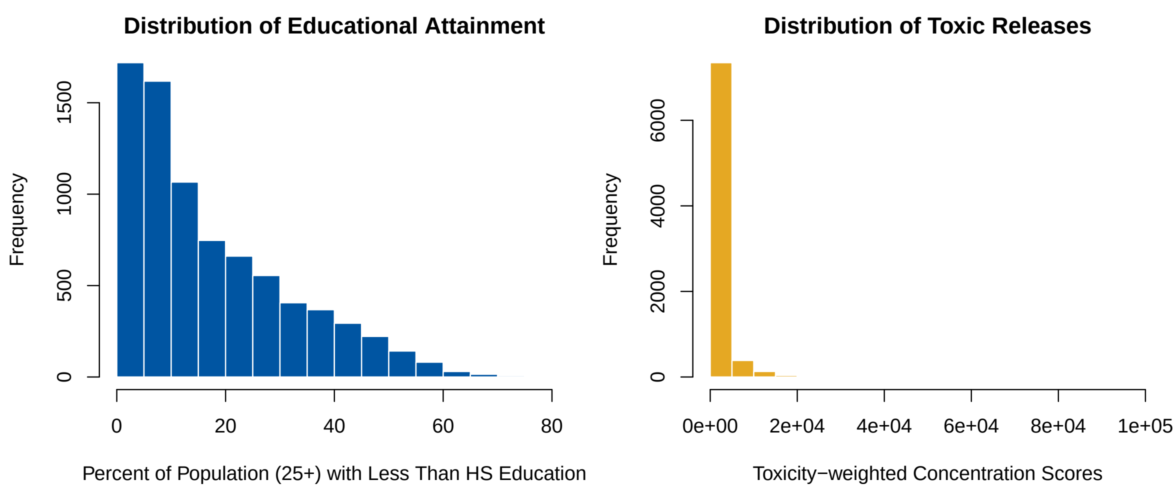

Figure 1: Distributions of Educational Attainment (left) and Toxic Releases (right).

3 Methods

A simple linear regression model was fitted to analyze the relationship between educational attainment and toxic release. The generalized equation for simple linear regression is presented below:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \quad \text{for } i = 1, \ldots, n$$

In this analysis, the response Y_i represents the toxic release variable and the predictor X_i represents the education variable. The intercept β_0 corresponds to the expected toxic release score in a census tract in California when the education variable equals zero. The slope parameter β_1 represents the expected change in toxic release for a one percentage point increase in the share of residents without a high school education. The random error term ε_i captures unexplained variance.

The linear regression model assumes validity, which is that the variables accurately reflect the quantities of interest and that the model includes all relevant predictors. In this dataset, both educational attainment percentages and toxic release scores are bounded and result from prior data processing, which may limit the precision of estimates. Another assumption of the linear regression model is that the data are representative of the population of interest. Representativeness will hold if inference is restricted to regions in California, but may be violated if extended to other areas.

For the error terms, the linear regression model assumes independence, constant variance, a mean of zero, and normality. The independence assumption requires that observations are not related to one another. In this research problem, nearby census tracts may be spatially correlated, meaning that educational attainment or toxic releases could be related across adjacent tracts. The assumption of homoscedasticity, or constant variance, means that the error terms have the same variance across all levels of the predictor variable. However, Figure 1 shows that the distribution of toxic release values is highly right-skewed, suggesting that the variance of residuals may increase for higher fitted values and that this assumption may not hold. Because the model is estimated using the ordinary least squares method to minimize the sum of squared residuals, the residuals have a sample mean of zero by construction, satisfying the assumption that the expected value of the error terms is zero. Finally, the model assumes that the error terms are normally distributed. Given the skewed distribution of the response variable, this assumption may be violated, but it can only be assessed after fitting the model and examining the residuals.

Because both variables are measured at the census tract level, this regression is ecological. The analysis describes associations between neighborhoods rather than individuals, so the results cannot be interpreted as showing that individuals without a high school degree are more exposed to toxic releases.

The linear regression was performed with the lm() function from the R programming language (R Core Team 2025).

4 Results

After training, the simple linear regression model produced model parameters of intercept $b_0 = 844.3$ and slope $b_1 = 42.2$. This implies that for a one percentage point increase in the proportion of residents without a high school education, the estimated toxic release score increases by about 42 units. The model explains only 2.9% of the variation in toxic release ($R^2 = 0.029$), indicating that educational attainment alone accounts for little of the variability in toxic release levels. Although the regression suggests a negative association between residents' high school educational attainment and toxic waste levels across California census tracts, the limited explanatory power of the model makes this relationship uncertain. If this relationship were confirmed, it would suggest that communities with lower educational attainment experience disproportionate exposure to environmental hazards, highlighting environmental inequality.

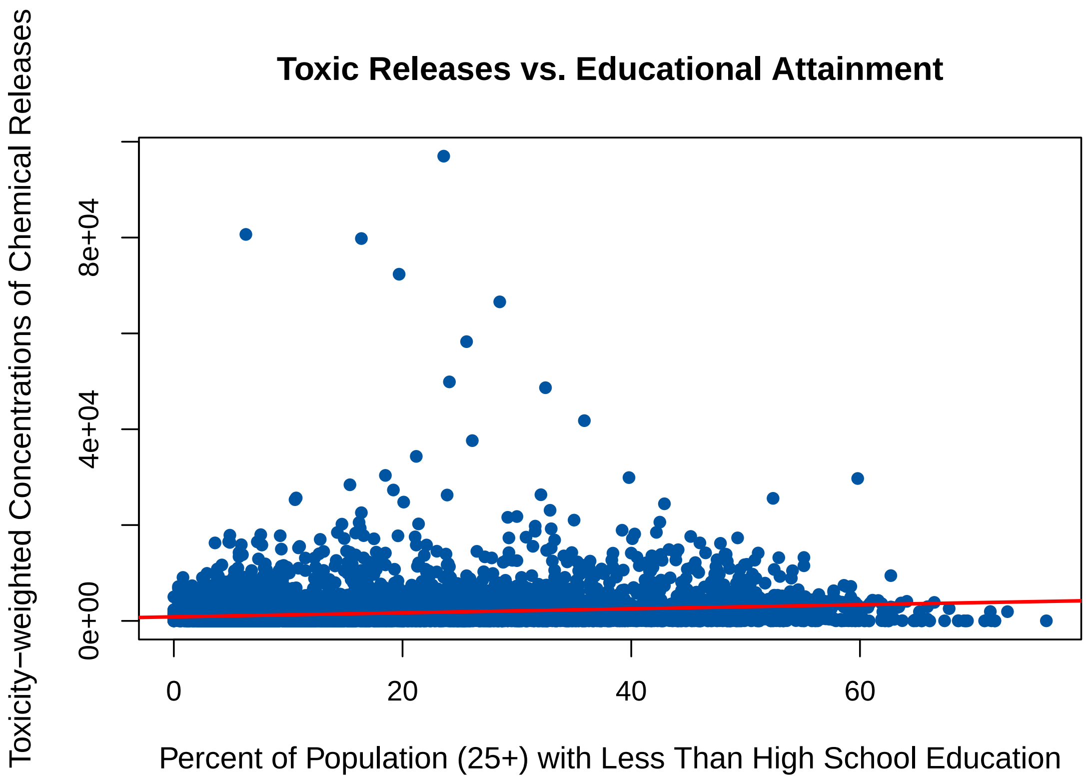

Figure 2: Scatter plot of Toxic Releases vs. Educational Attainment with fitted regression line shown in red.

These results are further illustrated in Figure 2. The scatter plot shows that most of the data is clustered near a toxic release variable level of 0. However, there are several massive outliers that may be affecting the regression line. The data is strongly right-skewed, and because the outcome spans several orders of magnitudes, the fitted regression line appears nearly flat.

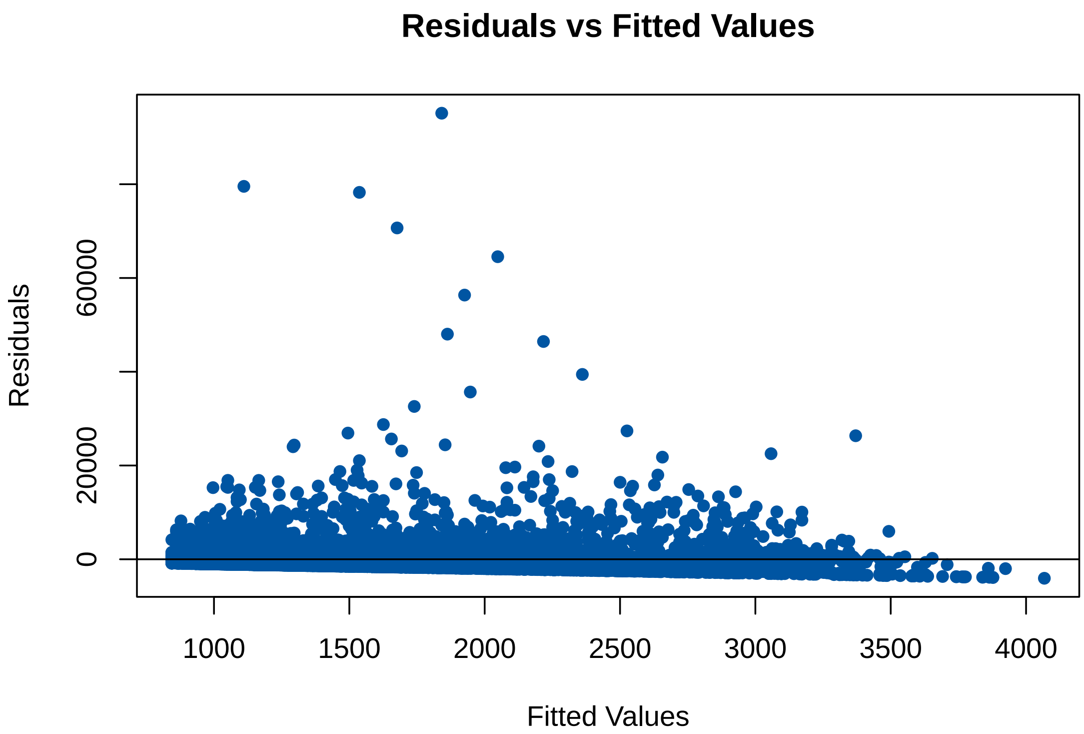

Figure 3: Residuals versus fitted values from the simple linear regression.

Plotting the residuals against the fitted values provides further insight into the validity of the regression assumptions. The residuals are not evenly scattered around the zero line and the spread of residuals decreases as fitted values increase. This pattern indicates heteroscedasticity, or unequal variance of the error terms, which violates the equal variance assumption of linear regression.

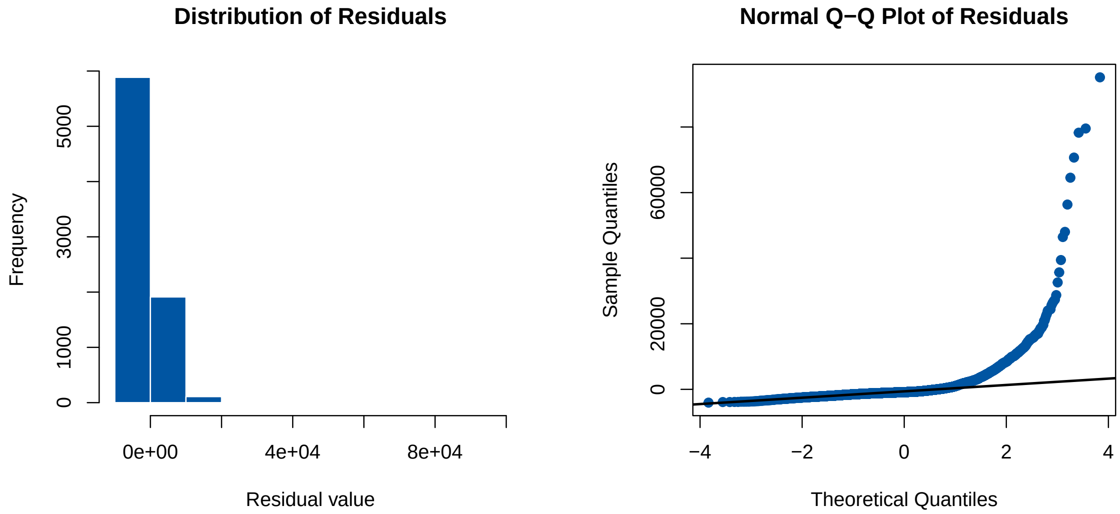

Figure 4: Histogram and Q-Q plot of residuals from the simple linear regression.

The histogram of residuals demonstrates a heavy right skew with extreme outliers, while the Q-Q plot shows the right tail deviating sharply from the reference line. Together, these diagnostics indicate a violation of the normality assumption for residuals in linear regression.

Although the simple linear regression model suggests that regions in California with lower educational attainment tend to experience higher levels of toxic release, the diagnostic plots reveal major violations of model assumptions, including non-normal residuals, unequal variance, and the influence of extreme outliers. These issues indicate that the linear model does not adequately capture the underlying relationship between education and toxic release. As a result, the findings cannot be interpreted as reliable evidence of an association. Overall, the analysis suggests that educational attainment alone explains little of the variation in toxic release, implying that toxic waste exposure is shaped by broader social and industrial factors. Future analyses could apply simple linear regression to log-transformed toxic release data to account for the right-skewed distribution. Alternatively, nonlinear or multiple regression models incorporating additional socioeconomic variables could be tested to better isolate the effect of education.

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